



```
1 . do "C:\Users\fahim\AppData\Local\Temp\STD962c_000000.tmp"
```

```
2 .
```

```
3 .
```

```
4 .
```

```
5 . /* HeartRate-with interaction */
```

```
6 . destring HeartRate, replace force
   HeartRate already numeric; no replace
```

```
7 .
```

```
8 . xtset Number TimeNumerical
```

Panel variable: **Number** (strongly balanced)

Time variable: **TimeNumerical**, 0 to 74, but with gaps

Delta: 1 unit

```
9 . xtmixed HeartRate GroupNumerical##TimeNumerical Sex weight || Number:, var reml
```

Performing EM optimization:

Performing gradient-based optimization:

Iteration 0: log restricted-likelihood = **-196.04968**

Iteration 1: log restricted-likelihood = **-196.04968**

Computing standard errors:

Mixed-effects REML regression

Group variable: **Number**

Number of obs = **52**

Number of groups = **18**

Obs per group:

min = **2**

avg = **2.9**

max = **3**

Wald chi2(10) = **110.23**

Prob > chi2 = **0.0000**

Log restricted-likelihood = **-196.04968**

HeartRate	Coefficient	Std. err.	z	P> z	[95% conf. interval]	
GroupNumerical						
1	-.7392634	13.91027	-0.05	0.958	-28.00288	26.52436
2	-4.544844	14.20488	-0.32	0.749	-32.3859	23.29621
TimeNumerical						
30	36.41454	9.95728	3.66	0.000	16.89863	55.93045
74	45.22415	9.95728	4.54	0.000	25.70824	64.74006
GroupNumerical#TimeNumerical						
1 30	-9.941936	14.54336	-0.68	0.494	-38.44641	18.56253
1 74	28.41807	14.08172	2.02	0.044	.8184039	56.01773
2 30	10.50645	14.08172	0.75	0.456	-17.09321	38.10612
2 74	3.515785	14.55	0.24	0.809	-25.0017	32.03327
Sex	-11.62464	10.28847	-1.13	0.259	-31.78967	8.540388
weight	.5029354	.9101197	0.55	0.581	-1.280866	2.286737
_cons	55.40788	71.54282	0.77	0.439	-84.81347	195.6292

Random-effects parameters	Estimate	Std. err.	[95% conf. interval]	
Number: Identity				
var(_cons)	277.4531	151.8359	94.92278	810.9775
var(Residual)	297.4423	79.31431	176.3705	501.6252

LR test vs. linear model: $\text{chibar2}(01) = 8.46$ Prob >= chibar2 = **0.0018**

10 . contrast TimeNumerical@GroupNumerical , effect

Contrasts of marginal linear predictions

Margins: **asbalanced**

	df	chi2	P>chi2
HeartRate			
TimeNumerical@GroupNumerical			
0	2	23.19	0.0000
1	2	55.92	0.0000
2	2	29.49	0.0000
Joint	6	108.61	0.0000

	Contrast	Std. err.	z	P> z	[95% conf. interval]	
HeartRate						
TimeNumerical@GroupNumerical						
(30 vs base) 0	36.41454	9.95728	3.66	0.000	16.89863	55.93045
(30 vs base) 1	26.47261	10.6001	2.50	0.013	5.696801	47.24841
(30 vs base) 2	46.921	9.95728	4.71	0.000	27.40509	66.43691
(74 vs base) 0	45.22415	9.95728	4.54	0.000	25.70824	64.74006
(74 vs base) 1	73.64222	9.95728	7.40	0.000	54.12631	93.15812
(74 vs base) 2	48.73993	10.6092	4.59	0.000	27.94628	69.53359

11 . contrast GroupNumerical@TimeNumerical , effect

Contrasts of marginal linear predictions

Margins: **asbalanced**

	df	chi2	P>chi2
HeartRate			
GroupNumerical@TimeNumerical			
0	2	0.11	0.9447
30	2	1.27	0.5305
74	2	5.08	0.0789
Joint	6	11.90	0.0641

	Contrast	Std. err.	z	P> z	[95% conf. interval]	
HeartRate						
GroupNumerical@TimeNumerical						
(1 vs base) 0	-.7392634	13.91027	-0.05	0.958	-28.00288	26.52436
(1 vs base) 30	-10.6812	14.36362	-0.74	0.457	-38.83337	17.47097
(1 vs base) 74	27.6788	13.91027	1.99	0.047	.4151832	54.94243
(2 vs base) 0	-4.544844	14.20488	-0.32	0.749	-32.3859	23.29621
(2 vs base) 30	5.961611	14.20488	0.42	0.675	-21.87945	33.80267
(2 vs base) 74	-1.029059	14.76941	-0.07	0.944	-29.97657	27.91845

12 .

13 .

14 .

15 .

16 . /* Repeat with natural log */

17 . generate logHeartRate = ln(HeartRate) /*natural log transform*/
(2 missing values generated)

18 . xtset Number TimeNumerical

Panel variable: **Number** (strongly balanced)Time variable: **TimeNumerical**, 0 to 74, but with gaps

Delta: 1 unit

19 . xtmixed logHeartRate GroupNumerical##TimeNumerical Sex weight || Number:, var residuals(ar, t(TimeNumerical)) n

Note: time gaps exist in the estimation data

Obtaining starting values by EM:

Performing gradient-based optimization:

Iteration 0: log restricted-likelihood = **3.4942185**Iteration 1: log restricted-likelihood = **3.4942187**

Computing standard errors:

Mixed-effects REML regression

Group variable: **Number**Number of obs = **52**Number of groups = **18**

Obs per group:

min = **2**avg = **2.9**max = **3**Wald chi2(10) = **145.92**Log restricted-likelihood = **3.4942187**Prob > chi2 = **0.0000**

logHeartRate	Coefficient	Std. err.	z	P> z	[95% conf. interval]	
GroupNumerical						
1	-.0044565	.1113229	-0.04	0.968	-.2226453	.2137323
2	-.0488643	.1138353	-0.43	0.668	-.2719774	.1742488
TimeNumerical						
30	.3253107	.0741536	4.39	0.000	.1799723	.4706491
74	.4106737	.0741536	5.54	0.000	.2653353	.5560121
GroupNumerical#TimeNumerical						
1 30	-.0706005	.1083803	-0.65	0.515	-.2830219	.141821
1 74	.1949524	.104869	1.86	0.063	-.0105872	.400492
2 30	.0954493	.104869	0.91	0.363	-.1100903	.3009888

2 74	.0302929	.1084219	0.28	0.780	-.1822102	.242796
Sex	-.0920075	.0851372	-1.08	0.280	-.2588733	.0748584
weight	.0044348	.0075213	0.59	0.555	-.0103067	.0191762
_cons	4.178087	.5908394	7.07	0.000	3.020063	5.336111

Random-effects parameters	Estimate	Std. err.	[95% conf. interval]	
Number: Identity				
var(_cons)	.0203002	.0101964	.007585	.0543304
Residual: AR(1)				
rho	-3.96e-09	.	.	.
var(e)	.0164963	5.26e-52	.0164963	.0164963

LR test vs. linear model: $\chi^2(2) = 11.51$ Prob > $\chi^2 = 0.0032$

Note: LR test is conservative and provided only for reference.

20 . contrast TimeNumerical@GroupNumerical , effect

Contrasts of marginal linear predictions

Margins: **asbalanced**

	df	chi2	P>chi2
logHeartRate			
TimeNumerical@GroupNumerical			
0	2	34.16	0.0000
1	2	67.18	0.0000
2	2	43.09	0.0000
Joint	6	144.43	0.0000

	Contrast	Std. err.	z	P> z	[95% conf. interval]	
logHeartRate						
TimeNumerical@GroupNumerical						
(30 vs base) 0	.3253107	.0741536	4.39	0.000	.1799723	.4706491
(30 vs base) 1	.2547102	.0790413	3.22	0.001	.0997921	.4096284
(30 vs base) 2	.42076	.0741536	5.67	0.000	.2754216	.5660984
(74 vs base) 0	.4106737	.0741536	5.54	0.000	.2653353	.5560121
(74 vs base) 1	.6056261	.0741536	8.17	0.000	.4602877	.7509645
(74 vs base) 2	.4409666	.0790984	5.57	0.000	.2859365	.5959966

21 . contrast GroupNumerical@TimeNumerical , effect

Contrasts of marginal linear predictions

Margins: **asbalanced**

	df	chi2	P>chi2
logHeartRate			
GroupNumerical@TimeNumerical			
0	2	0.22	0.8977
30	2	1.05	0.5915
74	2	3.97	0.1376
Joint	6	11.01	0.0880

	Contrast	Std. err.	z	P> z	[95% conf. interval]	
logHeartRate						
GroupNumerical@TimeNumerical						
(1 vs base) 0	-.0044565	.1113229	-0.04	0.968	-.2226453	.2137323
(1 vs base) 30	-.075057	.114539	-0.66	0.512	-.2995493	.1494354
(1 vs base) 74	.1904959	.1113229	1.71	0.087	-.0276929	.4086847
(2 vs base) 0	-.0488643	.1138353	-0.43	0.668	-.2719774	.1742488
(2 vs base) 30	.046585	.1138353	0.41	0.682	-.1765281	.2696981
(2 vs base) 74	-.0185714	.1178256	-0.16	0.875	-.2495053	.2123625

22 .

23 .

24 . /* HeartRate -without interaction */

25 . xtset Number TimeNumerical

Panel variable: **Number** (strongly balanced)

Time variable: **TimeNumerical**, 0 to 74, but with gaps

Delta: 1 unit

26 . xtmixed HeartRate Sex weight i.GroupNumerical i.TimeNumerical || Number:, var reml

Performing EM optimization:

Performing gradient-based optimization:

Iteration 0: log restricted-likelihood = -215.24338

Iteration 1: log restricted-likelihood = -215.24337

Computing standard errors:

Mixed-effects REML regression

Group variable: **Number**

Number of obs = 52

Number of groups = 18

Obs per group:

min = 2

avg = 2.9

max = 3

Wald chi2(6) = 80.09

Prob > chi2 = 0.0000

Log restricted-likelihood = -215.24337

HeartRate	Coefficient	Std. err.	z	P> z	[95% conf. interval]	
Sex	-12.10843	10.2933	-1.18	0.239	-32.28292	8.066061
weight	.4207657	.9109893	0.46	0.644	-1.364741	2.206272
GroupNumerical						
1	6.325318	11.35576	0.56	0.578	-15.93157	28.5822
2	.9715618	11.767	0.08	0.934	-22.09133	24.03445
TimeNumerical						
30	37.63184	6.510978	5.78	0.000	24.87056	50.39312
74	56.4235	6.512897	8.66	0.000	43.65845	69.18854
_cons	57.92167	71.40552	0.81	0.417	-82.03058	197.8739

Random-effects parameters	Estimate	Std. err.	[95% conf. interval]	
Number: Identity				
var(_cons)	254.4734	152.8005	78.43898	825.5679
var(Residual)	367.2763	91.60738	225.2595	598.8287

LR test vs. linear model: chibar2(01) = 6.37 Prob >= chibar2 = **0.0058**

27 .

28 . contrast TimeNumerical@GroupNumerical , effect

Contrasts of marginal linear predictions

Margins: **asbalanced**

	df	chi2	P>chi2
HeartRate			
TimeNumerical@GroupNumerical			
0	2	78.45	0.0000
1	2	78.45	0.0000
2	2	78.45	0.0000
Joint	2	78.45	0.0000

	Contrast	Std. err.	z	P> z	[95% conf. interval]	
HeartRate						
TimeNumerical@GroupNumerical						
(30 vs base) 0	37.63184	6.510978	5.78	0.000	24.87056	50.39312
(30 vs base) 1	37.63184	6.510978	5.78	0.000	24.87056	50.39312
(30 vs base) 2	37.63184	6.510978	5.78	0.000	24.87056	50.39312
(74 vs base) 0	56.4235	6.512897	8.66	0.000	43.65845	69.18854
(74 vs base) 1	56.4235	6.512897	8.66	0.000	43.65845	69.18854
(74 vs base) 2	56.4235	6.512897	8.66	0.000	43.65845	69.18854

29 . contrast GroupNumerical@TimeNumerical , effect

Contrasts of marginal linear predictions

Margins: **asbalanced**

	df	chi2	P>chi2
HeartRate			
GroupNumerical@TimeNumerical			
0	2	0.34	0.8427
30	2	0.34	0.8427
74	2	0.34	0.8427
Joint	2	0.34	0.8427

	Contrast	Std. err.	z	P> z	[95% conf. interval]	
HeartRate						
GroupNumerical@TimeNumerical						
(1 vs base) 0	6.325318	11.35576	0.56	0.578	-15.93157	28.5822
(1 vs base) 30	6.325318	11.35576	0.56	0.578	-15.93157	28.5822
(1 vs base) 74	6.325318	11.35576	0.56	0.578	-15.93157	28.5822
(2 vs base) 0	.9715618	11.767	0.08	0.934	-22.09133	24.03445
(2 vs base) 30	.9715618	11.767	0.08	0.934	-22.09133	24.03445
(2 vs base) 74	.9715618	11.767	0.08	0.934	-22.09133	24.03445

30 .

31 . /* Repeat with natural log */

32 . xtset Number TimeNumerical

Panel variable: **Number** (strongly balanced)

Time variable: **TimeNumerical**, 0 to 74, but with gaps

Delta: 1 unit

33 . xtmixed logHeartRate Sex weight i.GroupNumerical i.TimeNumerical || Number:, var reml

Performing EM optimization:

Performing gradient-based optimization:

Iteration 0: log restricted-likelihood = **4.2359597**

Iteration 1: log restricted-likelihood = **4.2359607**

Iteration 2: log restricted-likelihood = **4.2359607**

Computing standard errors:

Mixed-effects REML regression

Group variable: **Number**

Number of obs = **52**

Number of groups = **18**

Obs per group:

min = **2**

avg = **2.9**

max = **3**

Wald chi2(6) = **111.92**

Prob > chi2 = **0.0000**

Log restricted-likelihood = **4.2359607**

logHeartRate	Coefficient	Std. err.	z	P> z	[95% conf. interval]	
Sex	-.0955908	.0851704	-1.12	0.262	-.2625217	.0713401
weight	.0038187	.0075264	0.51	0.612	-.0109327	.0185701
GroupNumerical						
1	.0436587	.0938835	0.47	0.642	-.1403495	.2276669
2	-.0005957	.0971979	-0.01	0.995	-.1911	.1899087
TimeNumerical						
30	.3411822	.0480116	7.11	0.000	.2470813	.4352832
74	.4899538	.0480233	10.20	0.000	.3958298	.5840779
_cons	4.196232	.589842	7.11	0.000	3.040163	5.352302

Random-effects parameters	Estimate	Std. err.	[95% conf. interval]	
Number: Identity				
var(_cons)	.0191664	.0103412	.0066569	.0551831
var(Residual)	.0199516	.004975	.0122385	.0325257

LR test vs. linear model: $\text{chibar2}(01) = 9.47$ Prob >= chibar2 = 0.0010

34 .

35 . contrast TimeNumerical@GroupNumerical , effect

Contrasts of marginal linear predictions

Margins: **asbalanced**

	df	chi2	P>chi2
logHeartRate			
TimeNumerical@GroupNumerical			
0	2	110.43	0.0000
1	2	110.43	0.0000
2	2	110.43	0.0000
Joint	2	110.43	0.0000

	Contrast	Std. err.	z	P> z	[95% conf. interval]	
logHeartRate						
TimeNumerical@GroupNumerical						
(30 vs base) 0	.3411822	.0480116	7.11	0.000	.2470813	.4352832
(30 vs base) 1	.3411822	.0480116	7.11	0.000	.2470813	.4352832
(30 vs base) 2	.3411822	.0480116	7.11	0.000	.2470813	.4352832
(74 vs base) 0	.4899538	.0480233	10.20	0.000	.3958298	.5840779
(74 vs base) 1	.4899538	.0480233	10.20	0.000	.3958298	.5840779
(74 vs base) 2	.4899538	.0480233	10.20	0.000	.3958298	.5840779

36 . contrast GroupNumerical@TimeNumerical , effect

Contrasts of marginal linear predictions

Margins: **asbalanced**

	df	chi2	P>chi2
logHeartRate			
GroupNumerical@TimeNumerical			
0	2	0.27	0.8734
30	2	0.27	0.8734
74	2	0.27	0.8734
Joint	2	0.27	0.8734

	Contrast	Std. err.	z	P> z	[95% conf. interval]	
logHeartRate						
GroupNumerical@TimeNumerical						
(1 vs base) 0	.0436587	.0938835	0.47	0.642	-.1403495	.2276669
(1 vs base) 30	.0436587	.0938835	0.47	0.642	-.1403495	.2276669
(1 vs base) 74	.0436587	.0938835	0.47	0.642	-.1403495	.2276669
(2 vs base) 0	-.0005957	.0971979	-0.01	0.995	-.1911	.1899087
(2 vs base) 30	-.0005957	.0971979	-0.01	0.995	-.1911	.1899087
(2 vs base) 74	-.0005957	.0971979	-0.01	0.995	-.1911	.1899087

37 .

38 .

39 .

40 .

41 . /* CardiacOutput-with interaction */

42 . destring CardiacOutput, replace force

CardiacOutput already numeric; no **replace**

43 .

44 . xtset Number TimeNumerical

Panel variable: **Number** (strongly balanced)

Time variable: **TimeNumerical**, 0 to 74, but with gaps

Delta: 1 unit

45 . xtmixed CardiacOutput GroupNumerical##TimeNumerical Sex weight || Number:, var reml

Performing EM optimization:

Performing gradient-based optimization:

Iteration 0: log restricted-likelihood = -384.1288

Iteration 1: log restricted-likelihood = -384.1288

Computing standard errors:

Mixed-effects REML regression
Group variable: **Number**

Number of obs = **52**
Number of groups = **18**
Obs per group:
 min = **2**
 avg = **2.9**
 max = **3**
Wald chi2(10) = **60.00**
Prob > chi2 = **0.0000**

Log restricted-likelihood = **-384.1288**

CardiacOutput	Coefficient	Std. err.	z	P> z	[95% conf. interval]	
GroupNumerical						
1	1081.714	1413.57	0.77	0.444	-1688.832	3852.26
2	-1050.064	1445.241	-0.73	0.467	-3882.683	1782.556
TimeNumerical						
30	1079.114	950.1779	1.14	0.256	-783.2006	2941.428
74	3618.327	950.1779	3.81	0.000	1756.013	5480.642
GroupNumerical#TimeNumerical						
1 30	-776.2261	1388.64	-0.56	0.576	-3497.91	1945.458
1 74	1328.367	1343.754	0.99	0.323	-1305.343	3962.078
2 30	481.7082	1343.754	0.36	0.720	-2152.002	3115.419
2 74	-2637.57	1389.185	-1.90	0.058	-5360.322	85.18253
Sex	-479.0512	1076.924	-0.44	0.656	-2589.784	1631.682
weight	237.858	95.15307	2.50	0.012	51.36138	424.3546
_cons	-14591.78	7475.374	-1.95	0.051	-29243.24	59.68773

Random-effects parameters	Estimate	Std. err.	[95% conf. interval]	
Number: Identity				
var(_cons)	3224909	1722091	1132345	9184515
var(Residual)	2708514	734268	1592112	4607747

LR test vs. linear model: chibar2(01) = 9.07 Prob >= chibar2 = **0.0013**

46 . contrast TimeNumerical@GroupNumerical , effect

Contrasts of marginal linear predictions

Margins: **asbalanced**

	df	chi2	P>chi2
CardiacOutput			
TimeNumerical@GroupNumerical			
0	2	15.29	0.0005
1	2	32.99	0.0000
2	2	2.75	0.2530
Joint	6	51.03	0.0000

	Contrast	Std. err.	z	P> z	[95% conf. interval]	
CardiacOutput						
TimeNumerical@GroupNumerical						
(30 vs base) 0	1079.114	950.1779	1.14	0.256	-783.2006	2941.428
(30 vs base) 1	302.8877	1012.661	0.30	0.765	-1681.891	2287.667
(30 vs base) 2	1560.822	950.1779	1.64	0.100	-301.4924	3423.137
(74 vs base) 0	3618.327	950.1779	3.81	0.000	1756.013	5480.642
(74 vs base) 1	4946.694	950.1779	5.21	0.000	3084.38	6809.009
(74 vs base) 2	980.7573	1013.408	0.97	0.333	-1005.487	2967.001

47 . contrast GroupNumerical@TimeNumerical , effect

Contrasts of marginal linear predictions

Margins: **asbalanced**

	df	chi2	P>chi2
CardiacOutput			
GroupNumerical@TimeNumerical			
0	2	2.06	0.3563
30	2	0.34	0.8427
74	2	15.78	0.0004
Joint	6	19.94	0.0028

	Contrast	Std. err.	z	P> z	[95% conf. interval]	
CardiacOutput						
GroupNumerical@TimeNumerical						
(1 vs base) 0	1081.714	1413.57	0.77	0.444	-1688.832	3852.26
(1 vs base) 30	305.4882	1455.046	0.21	0.834	-2546.349	3157.326
(1 vs base) 74	2410.082	1413.57	1.70	0.088	-360.4643	5180.627
(2 vs base) 0	-1050.064	1445.241	-0.73	0.467	-3882.683	1782.556
(2 vs base) 30	-568.3553	1445.241	-0.39	0.694	-3400.975	2264.264
(2 vs base) 74	-3687.633	1496.722	-2.46	0.014	-6621.154	-754.1129

48 .

49 . /* Repeat with natural log */

50 . generate logCardiacOutput = ln(CardiacOutput) /*natural log transform*/
(2 missing values generated)

51 . xtset Number TimeNumerical

Panel variable: **Number** (strongly balanced)

Time variable: **TimeNumerical**, 0 to 74, but with gaps

Delta: 1 unit

52 . xtmixed logCardiacOutput GroupNumerical##TimeNumerical Sex weight || Number:, var residuals(ar, t(TimeNumerical
Note: time gaps exist in the estimation data

Obtaining starting values by EM:

Performing gradient-based optimization:

Iteration 0: log restricted-likelihood = -22.913371 (not concave)
Iteration 1: log restricted-likelihood = -22.913371 (not concave)
Iteration 2: log restricted-likelihood = -22.908831 (not concave)
Iteration 3: log restricted-likelihood = -22.764423
Iteration 4: log restricted-likelihood = -22.72614
Iteration 5: log restricted-likelihood = -22.695254
Iteration 6: log restricted-likelihood = -22.695073
Iteration 7: log restricted-likelihood = -22.695073

Computing standard errors:

Mixed-effects REML regression
Group variable: **Number**

Number of obs	=	52
Number of groups	=	18
Obs per group:		
min	=	2
avg	=	2.9
max	=	3
Wald chi2(10)	=	63.75
Prob > chi2	=	0.0000

Log restricted-likelihood = -22.695073

logCardiacOutput	Coefficient	Std. err.	z	P> z	[95% conf. interval]	
GroupNumerical						
1	.1187006	.2257383	0.53	0.599	-.3237384	.5611395
2	-.1910824	.2312522	-0.83	0.409	-.6443283	.2621635
TimeNumerical						
30	.1682361	.1261793	1.33	0.182	-.0790707	.4155429
74	.6476564	.1412145	4.59	0.000	.3708811	.9244316
GroupNumerical#TimeNumerical						
1 30	-.0325565	.184732	-0.18	0.860	-.3946245	.3295116
1 74	.0626793	.1997074	0.31	0.754	-.32874	.4540986
2 30	.1994848	.1784444	1.12	0.264	-.1502598	.5492295
2 74	-.3525769	.2065101	-1.71	0.088	-.7573293	.0521755
Sex	-.1030689	.1796602	-0.57	0.566	-.4551965	.2490586
weight	.0281465	.0158738	1.77	0.076	-.0029657	.0592586
_cons	6.081867	1.246075	4.88	0.000	3.639604	8.52413

Random-effects parameters	Estimate	Std. err.	[95% conf. interval]	
Number: Identity				
var(_cons)	.0898702	.0535324	.0279632	.2888318
Residual: AR(1)				
rho	.9509039	.0534968	.6330201	.9943794
var(e)	.0613023	.0263891	.0263667	.1425272

LR test vs. linear model: chi2(2) = 15.75 Prob > chi2 = 0.0004

Note: LR test is conservative and provided only for reference.

53 . contrast TimeNumerical@GroupNumerical , effect

Contrasts of marginal linear predictions

Margins: **asbalanced**

	df	chi2	P>chi2
logCardiacOutput			
TimeNumerical@GroupNumerical			
0	2	22.20	0.0000
1	2	27.54	0.0000
2	2	8.96	0.0113
Joint	6	58.71	0.0000

	Contrast	Std. err.	z	P> z	[95% conf. interval]	
logCardiacOutput						
TimeNumerical@GroupNumerical						
(30 vs base) 0	.1682361	.1261793	1.33	0.182	-.0790707	.4155429
(30 vs base) 1	.1356796	.1349248	1.01	0.315	-.1287681	.4001274
(30 vs base) 2	.3677209	.1261793	2.91	0.004	.1204141	.6150277
(74 vs base) 0	.6476564	.1412145	4.59	0.000	.3708811	.9244316
(74 vs base) 1	.7103357	.1412145	5.03	0.000	.4335604	.9871109
(74 vs base) 2	.2950794	.1506815	1.96	0.050	-.0002509	.5904097

54 . contrast GroupNumerical@TimeNumerical , effect

Contrasts of marginal linear predictions

Margins: **asbalanced**

	df	chi2	P>chi2
logCardiacOutput			
GroupNumerical@TimeNumerical			
0	2	1.72	0.4241
30	2	0.16	0.9226
74	2	9.25	0.0098
Joint	6	14.97	0.0205

	Contrast	Std. err.	z	P> z	[95% conf. interval]	
logCardiacOutput						
GroupNumerical@TimeNumerical						
(1 vs base) 0	.1187006	.2257383	0.53	0.599	-.3237384	.5611395
(1 vs base) 30	.0861441	.2306098	0.37	0.709	-.3658427	.5381309
(1 vs base) 74	.1813799	.2257383	0.80	0.422	-.2610591	.6238188
(2 vs base) 0	-.1910824	.2312522	-0.83	0.409	-.6443283	.2621635
(2 vs base) 30	.0084025	.2312522	0.04	0.971	-.4448435	.4616484
(2 vs base) 74	-.5436593	.2385651	-2.28	0.023	-1.011238	-.0760804

```

55 .
56 .
57 . /* CardiacOutput -without interaction */
58 . xtset Number TimeNumerical

```

Panel variable: **Number** (strongly balanced)
 Time variable: **TimeNumerical**, 0 to 74, but with gaps
 Delta: 1 unit

```
59 . xtmixed CardiacOutput Sex weight i.GroupNumerical i.TimeNumerical || Number:, var reml
```

Performing EM optimization:

Performing gradient-based optimization:

Iteration 0: log restricted-likelihood = **-422.90034**
 Iteration 1: log restricted-likelihood = **-422.90033**

Computing standard errors:

Mixed-effects REML regression	Number of obs	=	52
Group variable: Number	Number of groups	=	18
	Obs per group:		
	min =		2
	avg =		2.9
	max =		3
	Wald chi2(6)	=	35.87
Log restricted-likelihood = -422.90033	Prob > chi2	=	0.0000

CardiacOutput	Coefficient	Std. err.	z	P> z	[95% conf. interval]	
Sex	-531.0424	1111.316	-0.48	0.633	-2709.182	1647.098
weight	224.2798	98.23112	2.28	0.022	31.75034	416.8092
GroupNumerical						
1	1334.388	1225.184	1.09	0.276	-1066.928	3735.704
2	-1604.335	1268.63	-1.26	0.206	-4090.805	882.1346
TimeNumerical						
30	1069.935	640.1572	1.67	0.095	-184.7502	2324.62
74	3298.751	640.3197	5.15	0.000	2043.748	4553.755
_cons	-13392.18	7698.592	-1.74	0.082	-28481.15	1696.779

Random-effects parameters	Estimate	Std. err.	[95% conf. interval]	
Number: Identity				
var(_cons)	3212546	1855397	1035716	9964565
var(Residual)	3547567	900249.4	2157373	5833591

LR test vs. linear model: chibar2(01) = 7.13 Prob >= chibar2 = **0.0038**

60 .

61 . contrast TimeNumerical@GroupNumerical , effect

Contrasts of marginal linear predictions

Margins: **asbalanced**

	df	chi2	P>chi2
CardiacOutput			
TimeNumerical@GroupNumerical			
0	2	27.38	0.0000
1	2	27.38	0.0000
2	2	27.38	0.0000
Joint	2	27.38	0.0000

	Contrast	Std. err.	z	P> z	[95% conf. interval]	
CardiacOutput						
TimeNumerical@GroupNumerical						
(30 vs base) 0	1069.935	640.1572	1.67	0.095	-184.7502	2324.62
(30 vs base) 1	1069.935	640.1572	1.67	0.095	-184.7502	2324.62
(30 vs base) 2	1069.935	640.1572	1.67	0.095	-184.7502	2324.62
(74 vs base) 0	3298.751	640.3197	5.15	0.000	2043.748	4553.755
(74 vs base) 1	3298.751	640.3197	5.15	0.000	2043.748	4553.755
(74 vs base) 2	3298.751	640.3197	5.15	0.000	2043.748	4553.755

62 . contrast GroupNumerical@TimeNumerical , effect

Contrasts of marginal linear predictions

Margins: **asbalanced**

	df	chi2	P>chi2
CardiacOutput			
GroupNumerical@TimeNumerical			
0	2	4.94	0.0846
30	2	4.94	0.0846
74	2	4.94	0.0846
Joint	2	4.94	0.0846

	Contrast	Std. err.	z	P> z	[95% conf. interval]	
CardiacOutput						
GroupNumerical@TimeNumerical						
(1 vs base) 0	1334.388	1225.184	1.09	0.276	-1066.928	3735.704
(1 vs base) 30	1334.388	1225.184	1.09	0.276	-1066.928	3735.704
(1 vs base) 74	1334.388	1225.184	1.09	0.276	-1066.928	3735.704
(2 vs base) 0	-1604.335	1268.63	-1.26	0.206	-4090.805	882.1346
(2 vs base) 30	-1604.335	1268.63	-1.26	0.206	-4090.805	882.1346
(2 vs base) 74	-1604.335	1268.63	-1.26	0.206	-4090.805	882.1346

```

63 .
64 . /* Repeat with natural log */
65 . xtset Number TimeNumerical

Panel variable: Number (strongly balanced)
Time variable: TimeNumerical, 0 to 74, but with gaps
Delta: 1 unit

66 . xtmixed logCardiacOutput Sex weight i.GroupNumerical i.TimeNumerical || Number:, var reml

```

Performing EM optimization:

Performing gradient-based optimization:

```

Iteration 0: log restricted-likelihood = -25.172531
Iteration 1: log restricted-likelihood = -25.172531

```

Computing standard errors:

```

Mixed-effects REML regression
Group variable: Number
Number of obs      =      52
Number of groups   =      18
Obs per group:
    min =          2
    avg =          2.9
    max =          3
Wald chi2(6)       =      46.94
Prob > chi2        =      0.0000

Log restricted-likelihood = -25.172531

```

logCardiacOutput	Coefficient	Std. err.	z	P> z	[95% conf. interval]	
Sex	-.1075722	.1886249	-0.57	0.568	-.4772702	.2621258
weight	.0251036	.0166348	1.51	0.131	-.0075001	.0577072
GroupNumerical						
1	.1335841	.2076901	0.64	0.520	-.2734809	.5406492
2	-.2131438	.2147739	-0.99	0.321	-.6340929	.2078054
TimeNumerical						
30	.2332859	.0872615	2.67	0.008	.0622564	.4043153
74	.569343	.0872768	6.52	0.000	.3982836	.7404023
_cons	6.328178	1.303427	4.86	0.000	3.773508	8.882849

Random-effects parameters	Estimate	Std. err.	[95% conf. interval]	
Number: Identity				
var(_cons)	.1047673	.0515574	.0399335	.2748614
var(Residual)	.0658231	.0165764	.0401807	.1078298

LR test vs. linear model: chibar2(01) = 13.90 Prob >= chibar2 = 0.0001

67 .

68 . contrast TimeNumerical@GroupNumerical , effect

Contrasts of marginal linear predictions

Margins: asbalanced

	df	chi2	P>chi2
logCardiacOutput			
TimeNumerical@GroupNumerical			
0	2	42.82	0.0000
1	2	42.82	0.0000
2	2	42.82	0.0000
Joint	2	42.82	0.0000

	Contrast	Std. err.	z	P> z	[95% conf. interval]	
logCardiacOutput						
TimeNumerical@GroupNumerical						
(30 vs base) 0	.2332859	.0872615	2.67	0.008	.0622564	.4043153
(30 vs base) 1	.2332859	.0872615	2.67	0.008	.0622564	.4043153
(30 vs base) 2	.2332859	.0872615	2.67	0.008	.0622564	.4043153
(74 vs base) 0	.569343	.0872768	6.52	0.000	.3982836	.7404023
(74 vs base) 1	.569343	.0872768	6.52	0.000	.3982836	.7404023
(74 vs base) 2	.569343	.0872768	6.52	0.000	.3982836	.7404023

69 . contrast GroupNumerical@TimeNumerical , effect

Contrasts of marginal linear predictions

Margins: asbalanced

	df	chi2	P>chi2
logCardiacOutput			
GroupNumerical@TimeNumerical			
0	2	2.43	0.2963
30	2	2.43	0.2963
74	2	2.43	0.2963
Joint	2	2.43	0.2963

	Contrast	Std. err.	z	P> z	[95% conf. interval]	
logCardiacOutput						
GroupNumerical@TimeNumerical						
(1 vs base) 0	.1335841	.2076901	0.64	0.520	-.2734809	.5406492
(1 vs base) 30	.1335841	.2076901	0.64	0.520	-.2734809	.5406492
(1 vs base) 74	.1335841	.2076901	0.64	0.520	-.2734809	.5406492
(2 vs base) 0	-.2131438	.2147739	-0.99	0.321	-.6340929	.2078054
(2 vs base) 30	-.2131438	.2147739	-0.99	0.321	-.6340929	.2078054
(2 vs base) 74	-.2131438	.2147739	-0.99	0.321	-.6340929	.2078054

70 .

71 .

72 .

end of do-file

73 .